INLAND EMPIRE MEDICAL NETWORK, INC.
DERMATOLOGY AnhDao V. Le, M.D.

Topical Treatment for AK lesions

You have been diagnosed with **actinic keratosis** lesions (AKs). Treatment is important because untreated AKs may turn into a skin cancer called squamous cell carcinoma (SCC) Carac or Effudex or Diclofenac gel are all FDA approved for the topical treatment of multiple actinic or solar keratoses of the face or large diffused areas. These creams are not indicated for the treatment of SCC.

How should your treatment progress?

- ☐ *Apply* ***Carac or Effudex once daily in the evening, for ..... weeks****; Rest for ..... weeks. Then repeat cycle of treatment as instructed by your physician.*
- ☐ *Apply* ***Diclofenac gel twice daily, for ..... weeks.***

Individual results may vary. You may notice skin irritation and redness. This means the product is working. Irritation is localized to the skin area that is sundamaged and contains AK lesions. Your irritation may fade about 2 weeks after stopping Carac, Effudex, or Diclofenac.

Important Safety Information.

Carac or Effudex should not be used by women who are pregnant, considering becoming pregnant, or nursing. Carac® should not be used by people with dihydropyrimidine dehydrogenase (DPD) enzyme deficiency. The **most common side effects** are **skin reactions including redness, dryness, burning, pain, erosion** (loss of upper layer of skin), **swelling and eye irritation.**

How should I use this cream (Carac, Effudex, Diclofenac) ?

1. Clean the area where you will apply the cream. Rinse well and dry the area with a towel and wait 10 minutes before applying the cream.
2. Using your fingertips, put the cream sparingly but enough to cover the affected skin areas on your face, neck, chest, scalp, arms, hands, etc. as directed by your physician.
3. Wash your hands as soon as you finish putting the cream on your skin.
4. Avoid contact with your eyes, nostrils, and mouth.
5. A moisturizer/ broad spectrum sunscreen may be applied 2 hours after Carac or Effudex has been applied. Do not cover the treated skin with a dressing.
6. Do not use any other skin products including creams, lotions, medications or cosmetics, unless instructed by your physician.

What should I avoid while using this cream?

1. Avoid sunlight or other ultraviolet light (such as tanning booths) as much as possible. Sunlight may increase your side effects, including more sunburn, more reaction of redness, peeling, stinging. When exposed to sunlight, wear a wide-brimmed hat or umbrella and use UVA/UVB sunscreen with SPF 30+ or higher.
2. Do not breast feed or become pregnant while using Carac®. If you do become pregnant, stop using Carac or Effudex and tell your physician right away.

How can I prevent future AKs?

- Avoid excessive sunlight exposure, especially during peak hours (10 AM–4 PM)
- Use broad spectrum sunscreen against both UVA and UVB light, with a sun protection factor (SPF) of 30 or higher. Apply at least 20 minutes before sun exposure.
- Reapply sunscreen every 2 hours, even on cloudy days.
- When outside, wear a wide-brimmed hat and protective clothings to minimize sunlight exposure.

How do UVA & UVB Rays Affect Your Skin?

UVB rays are "B" Burning rays that penetrate at the epidermis level (superficial) causing acute redness, sunburn, and pigmentation of the skin (tan) by stimulating the melanin production. Its darkening effect is visible after 24 hours and is relatively long-lasting.

UVA rays are "A" Aging rays that penetrate pass the epidermis deeper into the dermis where the collagen and elastin reside, causing loss of elasticity resulting in prematurely aging skin, sagging or wrinkling skin. Its pigmentation effect is visible within one hour after sun-exposure but usually fades in 24 hours.

Too much lifetime exposure to both UVB & UVA rays can play a significant part in the formation of precancers called **actinic keratosis (AK)** and of skin cancers. The most common form of skin cancer is **Basal cell carcinoma (BCC)**, which appears as a pink, red or grayish flesh-toned pearly growth that persists more than a month. The second most common skin cancer is **Squamous cell carcinoma (SCC)**, which appears as a wart-like, crusty red growth that may bleed or persists as an open sore for many months. The third most common and most deadly form of skin cancer is a **melanoma (MM)**, which usually develops from a flat mole that has changes in 2-3 out of 5 ABCDEs.

A closer look

MODERATE SUN DAMAGE

- **Fine wrinkling:** Shallow indentations or lines, typically on the forehead and around the eyes and mouth
- **Roughness:** A change in skin texture from soft and supple to rough
- **Mottled coloring:** Light, patchy discoloration (hyperpigmentation) including brown spots

MORE SEVERE SUN DAMAGE

- **Coarse wrinkling:** Deep lines, furrows, and creases on the forehead, chin, and around the nose and eyes
- **Laxity:** Loose, thin skin that can result in a sunken appearance
- **Yellowing:** A change in skin tone from rosy to sallow or pale

Most of the sun damage signs can be improved by early sun protection practices, as well as by performing the numerous cosmetic and laser procedures available at your dermatologist or aesthetic cosmetic clinics.

Who is more likely at risk for AKs or skin cancers or getting early wrinkles and sun spots?

Those who have one or more of the following: **Pale** skin with **blond** or **red hair** and **light-colored eyes** (blue, hazel, green); a **lifetime history** of **frequent time in the sun** or **severe sunburns**; a tendency to **freckle or turn red** (burn) in the sun; a **weak immune system** or taking immunosuppressive medications; **over age 40**.

Early detection can save your life: Most moles are harmless, but it is important to recognize changes in a mole that might suggest a melanoma may be developing. Use the ABCDE rules to check for abnormal moles.

NORMAL MOLE

- Symmetrical shape
- Even coloration
- Even border

ABNORMAL MOLE

- **A is for asymmetrical:** One half of the mole does not match the other.
- **B is for border:** The mole has an irregular, ragged, or blurred edge.
- **C is for color:** The mole is multicolored (black, tan, gray, brown, and possibly blue, red, and white).
- **D is for diameter:** The mole is wider than a quarter inch or is smaller but growing.
- **E is for evolving:** An evolving mole is one that is changing in terms of size, shape, shades of color, or symptoms such as bleeding, itching, or tenderness.

You will need: a bright light, a full-length mirror, a hand mirror, two chairs or stools, and a blow-dryer.

1 Examine your scalp by using a blow-dryer to part hair.

2 Check your hands, including nails. In a full-length mirror, examine your elbows, arms, underarms, neck, chest, and torso. Women should also check under their breasts.

3 With your back to the mirror, use a hand mirror to inspect the back of your neck, shoulders, upper arms, back, buttocks, and legs.

4 Sitting down, check your legs and feet, including soles, heels, and nails.

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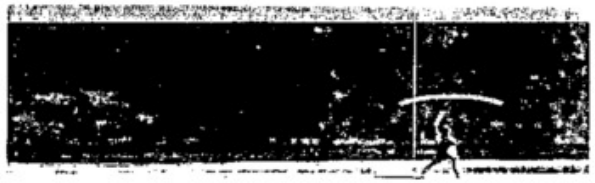

Safe and smart: a sun safety checklist

TO HELP PREVENT PHOTODAMAGE & SKIN CANCER

- ***Limit time in the intense sun.*** Try to avoid the hours when the sun rays are the strongest – 10 am to 4 pm.. Be sure to use high-SPF broad-spectrum sunscreen if you plan to be outside during these hours. Snow, water, ice, and reflective surfaces can make you burn faster. Don't get a sunburn or a suntan. Some sun exposure is healthy for getting the Vit –D.

- ***Cover up.*** Wear darker-colored clothes with tightly woven fabrics that cover your arms and legs and a wide-brimmed hat with neck covering. Use sun-protective clothing with built-in UPF (ultraviolet protective factor) 30 to 50+ .

- ***Apply sunscreen everyday!*** Use a broad-spectrum sunscreen (blocks UVA + UVB) with SPF 30 or higher; apply 15-30 minutes prior to sun-exposure; remember to reapply every 2 hours when you are active outdoors, and even more often if you are swimming, perspiring, or using a towel. Look in your sunscreen for the UVA photostabilizing complex (eg, Helioplex, Avo-triplex, Mexoryl, etc.) which will stabilize and maintain the chemical UVA protection from being breakdown within 30 minutes from intense sun exposure.

- ***No tanning bed is "safe".*** Tanning beds use less dangerous UVA rays, but these rays penetrate deep into your skin and increase your risk of skin cancer and AK, and alter your moles to become more atypical or turn into melanoma. Try using a self-tanner instead.

- ***Limit sun exposure when taking certain photo-sensitive medications.*** Discuss with your doctor whether any of your medications make you more sensitive to sunlight.

- ***Regularly exam your skin and get routine checkups with your doctor or dermatologist.*** Self-check your skin monthly for new skin growths or changes to existing freckles, moles, rough spots or bumps.